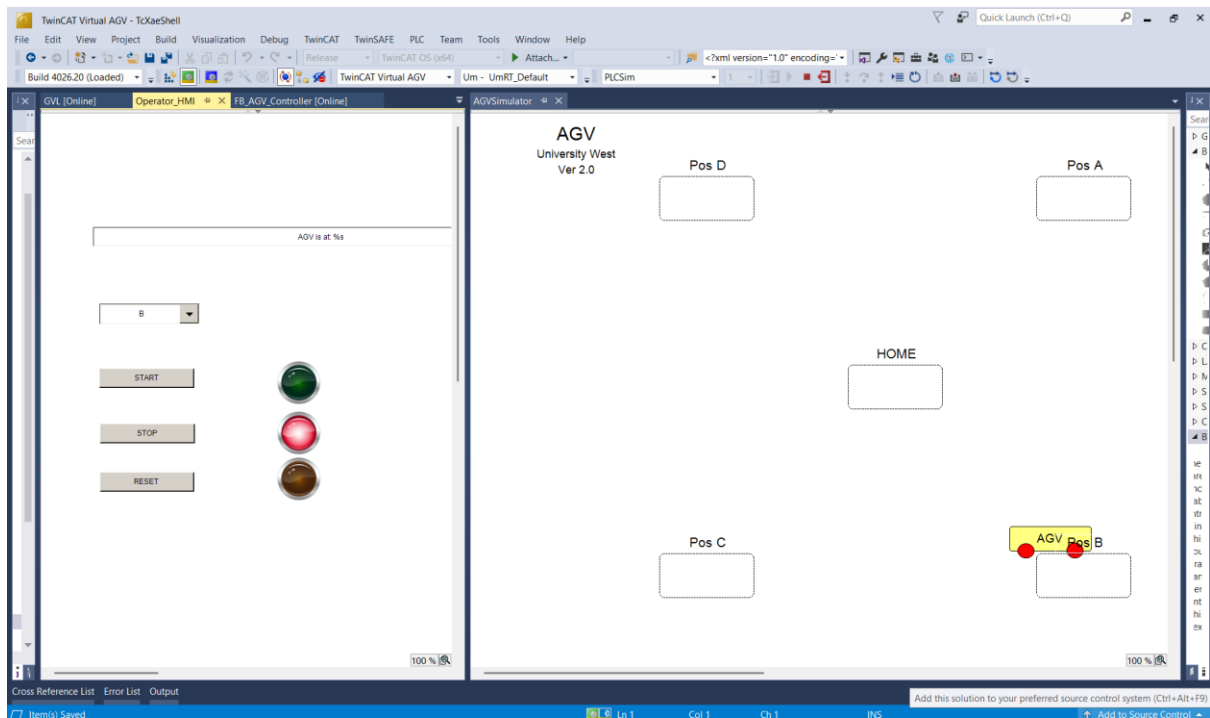




TwinCAT Virtual AGV - TcXaeShell

File Edit View Project Build Visualization Debug TwinCAT TwinSAFE PLC Team Tools Window Help

Build 4026.20 (Loaded) TwinCAT Virtual AGV Um - UmRT_Default PLCSim

GVL [Online] Operator_HMI FB_AGV_Controller [Online] GlobalTextList TL_Targets GVL Process

TwinCAT_Virtual_AGV.PLCsim.MAIN.myAGVController

Expression	Type	Value	Prepared value	Address	Comment
iTargetSelect	INT	3			From Dropdown: 1=Home, 2=A, 3...
bStart	BOOL	FALSE			
bStop	BOOL	FALSE			
bReset	BOOL	FALSE			
bAGVStrobeIn	BOOL	FALSE			Connect to GVL.qxAGVStrobe
iAGVAtPos	INT	0			Connect to GVL.qiAtPos
bReadyInd	BOOL	FALSE			Lamp: Ready to start
bMovingInd	BOOL	FALSE			Lamp/Animation: AGV is moving
iPositionDisp	INT	0			Text Display: Act...Position (0 = Off)
bStopInd	BOOL	TRUE			Lamp: Immediate stop active
bPLCStrobeOut	BOOL	FALSE			Connect to GVL.ixPLCStrobe
iPLCGotoPos	INT	0			Connect to GVL.iiGotoPos
iState	INT	100			State Machine step
fbStartTrig	R_TRIG				Edge triggers for buttons to preven...
fbStopTrig	R_TRIG				
fbResetTrig	R_TRIG				

```
29
30 1: // DELAY 1 CYCLE FOR AGV TO REGISTER TARGET
31   bPLCStrobeOut[FALSE] := TRUE; // NOW raise the strobe
32   iState[100] := 10;
33
34 10: // WAIT FOR ORDER RECEIVED
35   IF bAGVStrobeIn[FALSE] AND iAGVAtPos[0] = 0 THEN
36     bPLCStrobeOut[FALSE] := FALSE;
37     iState[100] := 20;
38   END_IF
39
40 20: // WAIT FOR AGV TO START MOVING
41   IF NOT bAGVStrobeIn[FALSE] THEN
42     bMovingInd[FALSE] := TRUE;
43     iState[100] := 30;
44   END_IF
45
46 30: // MOVING... WAIT FOR ANSWER
47   IF bAGVStrobeIn[FALSE] AND iAGVAtPos[0] > 0 THEN
48     bMovingInd[FALSE] := FALSE;
49     iPositionDisp[0] := iAGVAtPos[0];
50     iPLCGotoPos[0] := 0;
51     iState[100] := 31;
52   END_IF
```

Cross Reference List Error List Output

Item(s) Saved 0 Ln 1

TwinCAT Virtual AGV - TcXaeShell

File Edit View Project Build Visualization Debug TwinCAT TwinSAFE PLC Team Tools Window Help

Release TwinCAT OS (x64) Attach... <?xml version="1.0" encoding="UTF-8" ?>

Build 4026.20 (Loaded) TwinCAT Virtual AGV Um - UmRT_Default PLCSim 1

Solution Explorer

Search Solution Explorer (Ctrl+)

- License
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 - System_VisuElems
 - System_VisuElemsSpecialConti
 - System_VisuElemsWinControls
 - System_VisuElemTextEditor
 - system_visuinputs
 - System_VisuNativeControl
 - Tc2_Standard
 - Tc2_System
 - Tc3_Module
 - VisuSymbols
 - DUTs
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 - AGV (FB)
 - FB_AGV_Controller (FB)
 - MAIN (PRG)
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 - VISUs
 - AGVSimulator
 - Operator_HMI
 - TL_Targets
 - GlobalTextList

GVL [Online] Operator_HMI FB_AGV_Controller [Online] GlobalTextList TL_Targets

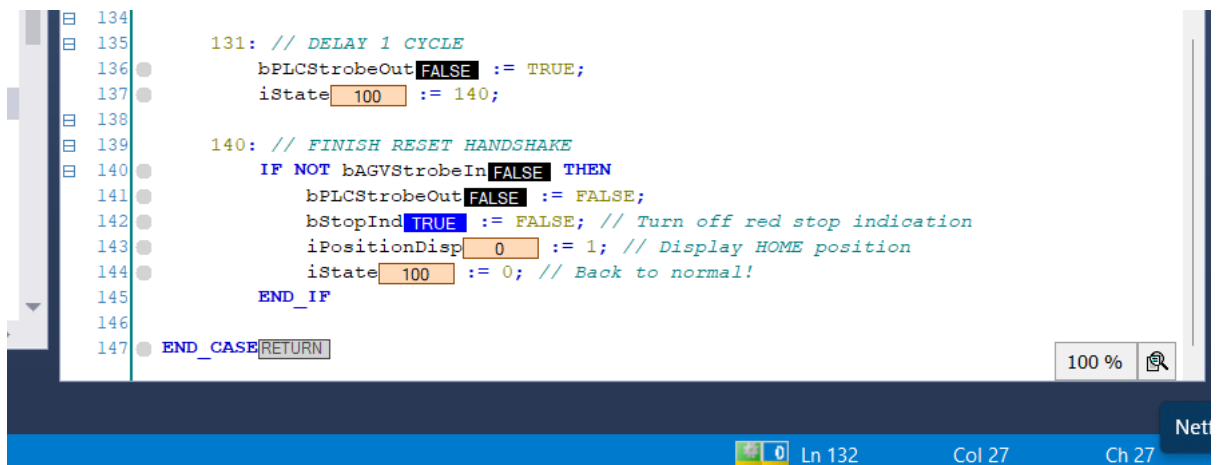
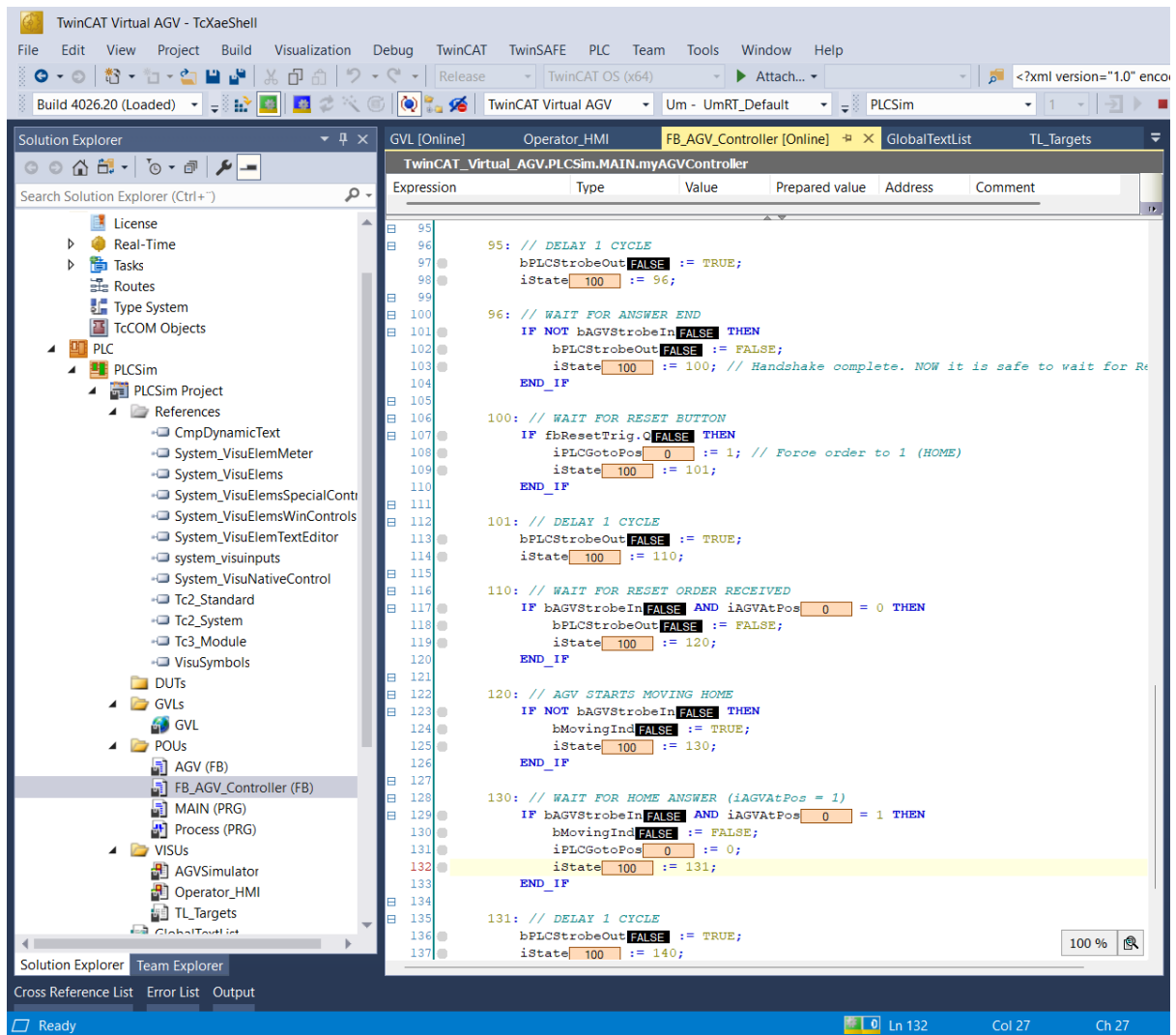
TwinCAT_Virtual_AGV.PLCSim.MAIN.myAGVController

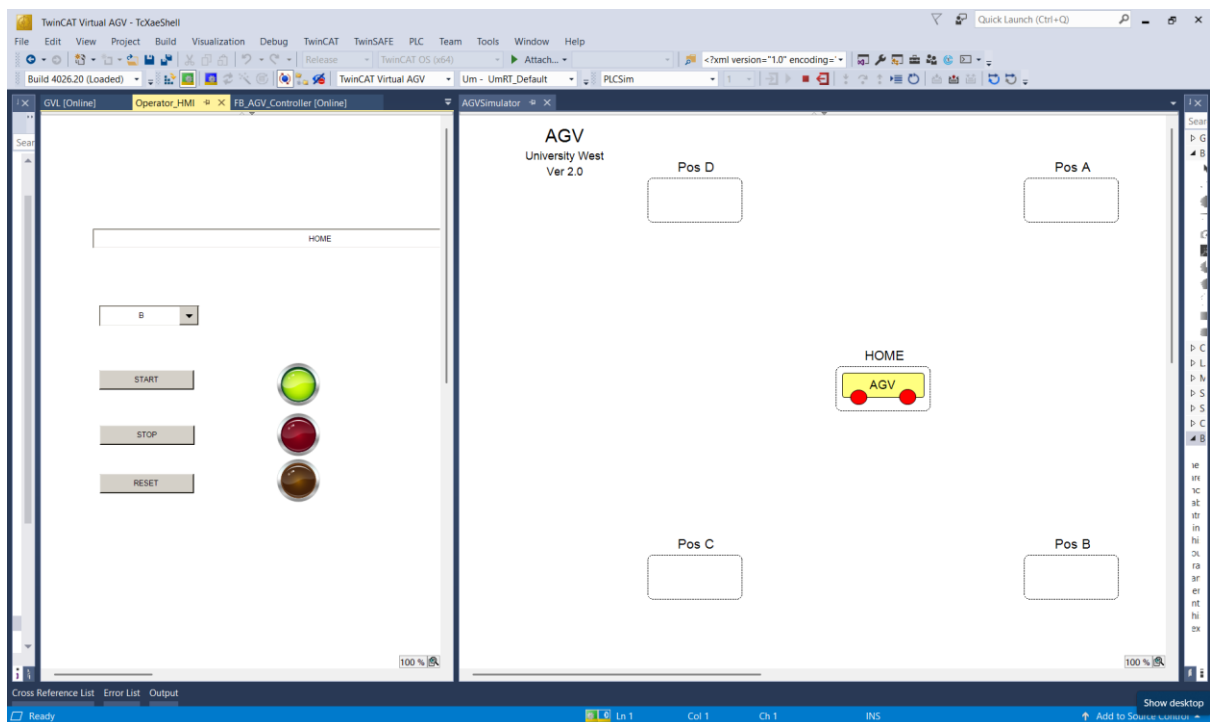
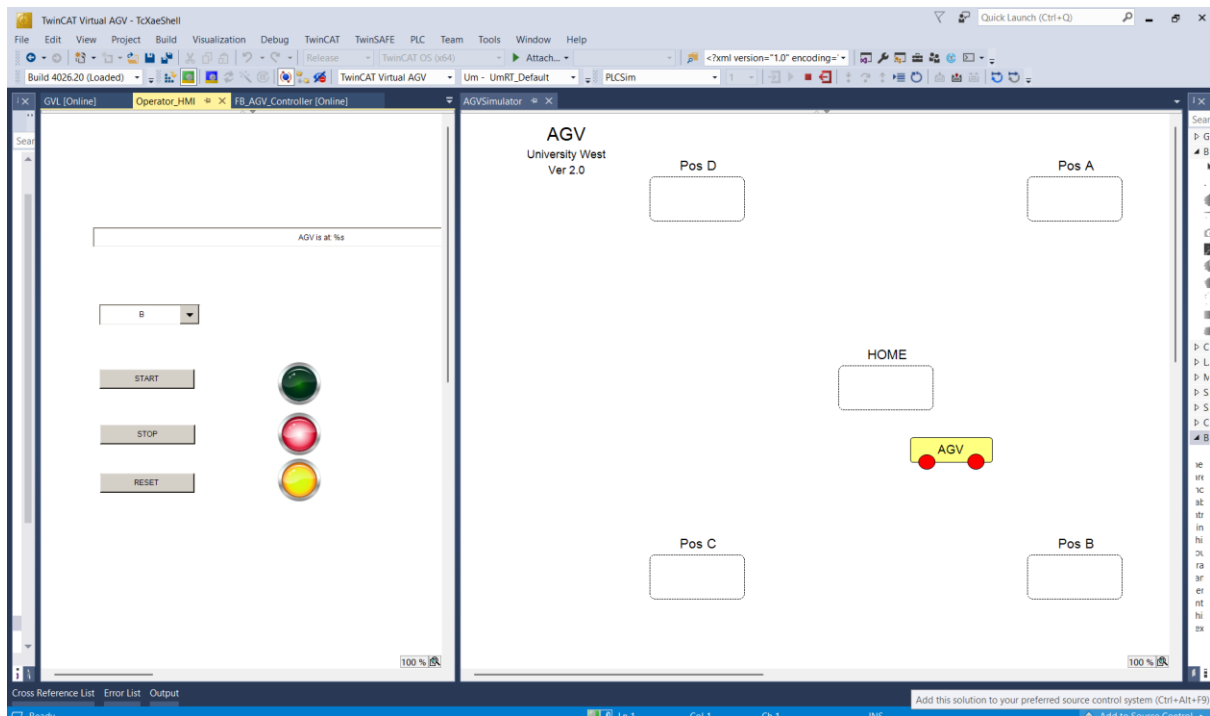
Expression	Type	Value	Prepared value	Address	Comment
------------	------	-------	----------------	---------	---------

```
53  
54 31: // DELAY 1 CYCLE  
55 bPLCStrobeOut FALSE := TRUE;  
56 iState 100 := 40;  
57  
58 40: // WAIT FOR ANSWER END  
59 IF NOT bAGVStrobeIn FALSE THEN  
60 bPLCStrobeOut FALSE := FALSE;  
61 iState 100 := 0;  
62 END_IF  
63  
64 // -----  
65 // STOP AND RESET SEQUENCE (Fully Complete Handshake)  
66 // -----  
67 90: // INITIATE STOP COMMAND  
68 iPLCGotoPos 0 := 6; // Command 6: Stop Instantly  
69 bMovingInd FALSE := FALSE;  
70 bStopInd TRUE := TRUE;  
71 iPositionDisp 0 := 0;  
72 iState 100 := 91;  
73  
74  
75 91: // DELAY 1 CYCLE  
76 bPLCStrobeOut FALSE := TRUE;  
77 iState 100 := 92;  
78  
79 92: // WAIT FOR STOP ORDER RECEIVED (AGV Ack)  
80 IF bAGVStrobeIn FALSE THEN  
81 bPLCStrobeOut FALSE := FALSE;  
82 iState 100 := 93;  
83 END_IF  
84  
85 93: // WAIT FOR AGV TO STOP AND SEND ANSWER  
86 IF NOT bAGVStrobeIn FALSE THEN  
87 iState 100 := 94;  
88 END_IF  
89  
90 94: // WAIT FOR AGV "STOPPED" ANSWER (iAGVatPos = 6)  
91 IF bAGVStrobeIn FALSE AND iAGVatPos 0 = 6 THEN  
92 iPLCGotoPos 0 := 0; // Acknowledge we heard the answer  
93 iState 100 := 95;  
94 END_IF  
95  
96 95: // DELAY 1 CYCLE
```

100 %

Ready Ln 1 Col 1 Ch 1





FB_AGV_Controller :

// Read HMI button presses (Rising Edges)

fbStartTrig(CLK := bStart);

fbStopTrig(CLK := bStop);

fbResetTrig(CLK := bReset);

```

// -----
// PRIORITY 1: IMMEDIATE STOP HANDLING
// If Stop is pressed while moving/starting (States 10 to 40)
// -----
IF fbStopTrig.Q AND iState >= 10 AND iState <= 40 THEN
    iState := 90; // Jump immediately to Stop Sequence
END_IF

// -----
// MAIN STATE MACHINE
// -----
CASE iState OF

    0: // IDLE / READY STATE
        bReadyInd := TRUE;
        bMovingInd := FALSE;
        IF fbStartTrig.Q AND iTargetSelect >= 1 AND iTargetSelect <= 5 THEN
            bReadyInd := FALSE;
            iPositionDisp := 0;

            // Set order FIRST
            iPLCGotoPos := iTargetSelect;
            iState := 1; // Go to intermediate step
        END_IF

    1: // DELAY 1 CYCLE FOR AGV TO REGISTER TARGET
        bPLCStrobeOut := TRUE; // NOW raise the strobe
        iState := 10;

```

10: // WAIT FOR ORDER RECEIVED

IF bAGVStrobeIn AND iAGVAtPos = 0 THEN

bPLCStrobeOut := FALSE;

iState := 20;

END_IF

20: // WAIT FOR AGV TO START MOVING

IF NOT bAGVStrobeIn THEN

bMovingInd := TRUE;

iState := 30;

END_IF

30: // MOVING... WAIT FOR ANSWER

IF bAGVStrobeIn AND iAGVAtPos > 0 THEN

bMovingInd := FALSE;

iPositionDisp := iAGVAtPos;

iPLCGotoPos := 0;

iState := 31;

END_IF

31: // DELAY 1 CYCLE

bPLCStrobeOut := TRUE;

iState := 40;

40: // WAIT FOR ANSWER END

IF NOT bAGVStrobeIn THEN

bPLCStrobeOut := FALSE;

iState := 0;

END_IF

// -----

// STOP AND RESET SEQUENCE (Fully Complete Handshake)

// -----

90: // INITIATE STOP COMMAND

 iPLCGotoPos := 6; // Command 6: Stop Instantly

 bMovingInd := FALSE;

 bStopInd := TRUE;

 iPositionDisp := 0;

 iState := 91;

91: // DELAY 1 CYCLE

 bPLCStrobeOut := TRUE;

 iState := 92;

92: // WAIT FOR STOP ORDER RECEIVED (AGV Ack)

 IF bAGVStrobeIn THEN

 bPLCStrobeOut := FALSE;

 iState := 93;

 END_IF

93: // WAIT FOR AGV TO STOP AND SEND ANSWER

 IF NOT bAGVStrobeIn THEN

 iState := 94;

 END_IF

94: // WAIT FOR AGV "STOPPED" ANSWER (iAGVAtPos = 6)

IF bAGVStrobeIn AND iAGVAtPos = 6 THEN

iPLCGotoPos := 0; // Acknowledge we heard the answer

iState := 95;

END_IF

95: // DELAY 1 CYCLE

bPLCStrobeOut := TRUE;

iState := 96;

96: // WAIT FOR ANSWER END

IF NOT bAGVStrobeIn THEN

bPLCStrobeOut := FALSE;

iState := 100; // Handshake complete. NOW it is safe to wait for Reset.

END_IF

100: // WAIT FOR RESET BUTTON

IF fbResetTrig.Q THEN

iPLCGotoPos := 1; // Force order to 1 (HOME)

iState := 101;

END_IF

101: // DELAY 1 CYCLE

bPLCStrobeOut := TRUE;

iState := 110;

110: // WAIT FOR RESET ORDER RECEIVED

IF bAGVStrobeIn AND iAGVAtPos = 0 THEN

```
    bPLCStrobeOut := FALSE;

    iState := 120;

END_IF
```

120: // AGV STARTS MOVING HOME

```
IF NOT bAGVStrobeIn THEN

    bMovingInd := TRUE;

    iState := 130;

END_IF
```

130: // WAIT FOR HOME ANSWER (iAGVAtPos = 1)

```
IF bAGVStrobeIn AND iAGVAtPos = 1 THEN

    bMovingInd := FALSE;

    iPLCGotoPos := 0;

    iState := 131;

END_IF
```

131: // DELAY 1 CYCLE

```
    bPLCStrobeOut := TRUE;

    iState := 140;
```

140: // FINISH RESET HANDSHAKE

```
IF NOT bAGVStrobeIn THEN

    bPLCStrobeOut := FALSE;

    bStopInd := FALSE; // Turn off red stop indication

    iPositionDisp := 1; // Display HOME position

    iState := 0; // Back to normal!

END_IF
```

END_CASE

TwinCAT Virtual AGV - TcXaeShell

FileEditViewProjectBuildVisualizationTextlistDebugTwinCATTwinSAFEPLCTeamToolsWindowHelp

Build 4026.20 (Loaded)

TwinCAT Virtual AGV

Um - UmRT_Default

PLC...

Release

TwinCAT OS (x64)

Attach...

AGV [Online]GVL [Online]Operator_HMIGlobalTextListFB_AGV_Controller [Online]

Search

ID	Default
546	AGV
699	AGV is at: %s
738	ERROR
838	HOME
50	MOVING
505	Pos A
284	Pos AD
908	Pos B
412	Pos C
547	Pos D
54	READY
917	RESET
664	START
733	STOP
272	STOP ACTIVE
315	University West
428	Ver 2.0

Cross Reference ListError ListOutput

Ready

TwinCAT Virtual AGV - TcXaeShell

FileEditViewProjectBuildVisualizationTextlistDebugTwinCATTwinSAFEPLCTeamToolsWindowHe

Build 4026.20 (Loaded)

ReleaseTwinCAT OS (x64)Attach...

TwinCAT Virtual AGVUm - UmRT_DefaultPLCS

Process [Online]AGV [Online]GVL [Online]GlobalTextListTL_Targets

Search

ID	Default
1	HOME
2	A
3	B
4	C
5	D
6	STOPPED

Cross Reference ListError ListOutput

Ready

TwinCAT Virtual AGV - TcXaeShell

File Edit View Project Build Visualization FBD/LD/IL Debug TwinCAT TwinSAFE PLC Team Tools Window Help

Build 4026.20 (Loaded) TwinCAT Virtual AGV Um - UmRT_Default PLCSim

GVL [Online] Operator_HMI Process [Online] GlobalTextList TL_Targets AGV [Online]

TwinCAT_Virtual_AGV.PLC.Sim.Process

Expression	Type	Value	Prepared value	Address	Comment
fbAGV	AGV				

Search

1

GVL.ixPLCStrobe

fbAGV

AGV

xStrobeIn

xStrobeOut

GVL.qxAGVStrobe

GVL.iiGoToPos 0

iGoToPos

iAtPos

GVL.qiAtPos 0

iSpeed 1

RET

100 %

Cross Reference List Error List Output

Ready

Ln 1

Git